

Measuring Visible Lines in the Hydrogen Spectrum

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In this experiment, students use light dispersion and diffraction to measure the wavelengths of spectral lines. Reliable results require careful calibration of the device, and the basic principles of calibration are briefly introduced. The data analysis involves propagating measurement uncertainties and applying weighted regression to obtain accurate wavelength values. The experiment shows how simple measurements can reveal the quantum nature of matter.

Introduction

When matter is heated or exposed to external energy sources such as electric discharge or cathode rays, it begins to emit electromagnetic radiation. This radiation is not uniform but consists of waves of different wavelengths, forming what is known as a **spectrum**. Spectra can be studied by using dispersion and diffraction, which separate the radiation into its wavelength components.

Different states of matter produce different types of spectra. Hot solids and liquids emit radiation across almost all wavelengths, forming a **continuous spectrum** — seen as a smooth band of colors from red to violet. Heated gases and vapors, in contrast, produce either **band spectra** (broad colored bands separated by dark gaps) or **line spectra** (sharp, discrete lines). Thus, the type of spectrum depends strongly on the physical state of the substance.

In solids, atoms are arranged in a rigid lattice due to strong interactions. In liquids and amorphous solids, interactions are weaker because of greater atomic motion. In gases and vapors, atoms and molecules are far less bound, so their emission spectra consist of distinct lines that directly reflect the internal structure of the atoms.

The hydrogen atom, being the simplest of all, provides the clearest example of such a line spectrum. Its study opened the way to modern atomic physics and serves as the foundation for this laboratory experiment.

Hydrogen Spectrum Empirical Models

The simplest atomic spectrum is that of hydrogen, the lightest chemical element. In 1885 ¹ Johann Balmer studied its visible lines and discovered a simple mathematical regularity in their wavelengths. He expressed these wavelengths λ as

$$\lambda = \lambda_0 \cdot \frac{n^2}{n^2 - 4},$$

¹ Johann Jakob Balmer. Notiz über die spectrallinien des wasserstoffs. *Annalen der Physik und Chemie*, 25:80–87, 1885

where $\lambda_0 = 364.56 \text{ nm}$ (Balmer's constant), and n takes integer values $3, 4, 5, 6, \dots$

The lines described by this relation are known as the **Balmer series**. In the visible spectrum, Balmer identified four strong lines.

Later, further study revealed that all of hydrogen's spectral lines could be described more generally by the **Rydberg formula**:²

$$\frac{1}{\lambda} = R \left(\frac{1}{m^2} - \frac{1}{n^2} \right) \quad (1)$$

where m and n are positive integers with $n > m$, and $R = 109\,721.6 \text{ cm}^{-1}$ is the Rydberg constant.

For $m = 1$ the formula gives **Lyman** series in the ultraviolet. For $m = 2$ it reproduces the **Balmer** series. For $m = 3, 4, 5$ it yields the **Paschen**, **Brackett** and **Pfund** series respectively, lying in the infrared.

Expressing the relation in terms of frequency $\nu = c/\lambda$ gives:

$$\nu = \frac{cR}{m^2} - \frac{cR}{n^2} = T_m - T_n, \quad (2)$$

where $T_k = \frac{cR}{k^2}$ is called a **spectral term**. The frequency of each spectral line can thus be represented as the difference between two terms, a structure that hints at the quantized energy levels within the atom.

The empirical models of Balmer and Rydberg provided remarkably accurate predictions for hydrogen's spectrum, even before the physical nature of these spectral terms was understood. Their success paved the way for later theoretical explanations based on the quantum structure of atoms.

Bohr's Theory of the Hydrogen Atom

Early models treated atoms as tiny oscillators that emit electromagnetic waves when their charges move. While this idea could explain some optical effects, it could not predict the specific frequencies observed in atomic spectra. Experiments showed that the charge-to-mass ratio of the "oscillating particle" matched that of the electron, suggesting that light emission is tied to the motion of electrons inside atoms.

Ernest Rutherford's experiments with α -particle scattering on gold foil led him to propose a **planetary model** of the atom: a dense, positively charged nucleus at the center with electrons orbiting around it (Figure 1).

According to this model, the center of the atom is the positively charged nucleus with a diameter of the order of $10^{-15} - 10^{-14} \text{ m}$, where almost the entire mass of the atom is concentrated. The charge of the nucleus $q = eZ$, where Z is the number of the element in the periodic table, e is the value of the electron charge. Since the atom as

² Johannes Robert Rydberg. On the structure of the line-spectra of the chemical elements (preliminary notice). *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 29 (179):331–337, 1890. doi: 10.1080/14786449008619945.

Here $c = 299\,792\,458 \text{ m/s}$ is speed of light.

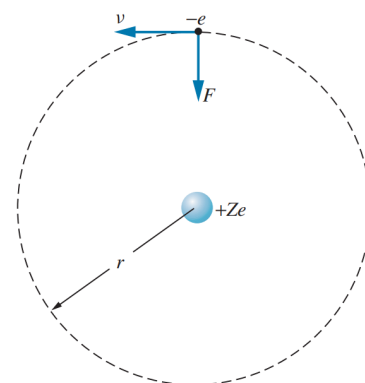


Figure 1: Atom planetary model ($Z = 1$ for hydrogen). Image: [Krane(1996), p.183].

a whole is neutral, the number of electrons in it is equal to Z . Electrons interact with the nucleus and with each other by electric forces according to Coulomb's law.

In this model, the electron's motion should determine the frequency of emitted radiation. However, classical physics predicts that an accelerating electron continuously radiates energy, spirals into the nucleus, and the atom collapses in a fraction of a second (it is of order 10^{-12} sec, as shown in textbooks³). Moreover, the model allows orbits of any size and energy, while experiments show that atoms emit only discrete spectral lines. Clearly, a new idea was needed.

To overcome this difficulty, Niels Bohr in 1913 combined Planck's idea of quantized energy with Rutherford's planetary model and introduced postulates that explained atomic stability and predicted hydrogen's spectrum. Here is what he wrote in his revolutionary article "On the constitution of atoms and molecules" [Bohr(1913)]:

💡 Bohr's postulates:

1. "The dynamical equilibrium of the systems in the stationary states can be discussed by help of the ordinary mechanics, while the passing of the systems between different stationary states cannot be treated on that basis".
2. "The latter process is followed by the emission of a homogeneous radiation, for which the relation between the frequency and the amount of energy emitted is the one given by Planck's theory".

The first postulate introduces the idea of the **stationary state**. The second postulate describes **photon emission and absorption** phenomena that can be written as

$$\Delta E = h\nu = E_{n_1} - E_{n_2},$$

where h is Planck's constant, and ν is photon frequency (see Figure 2).

Further in this article Bohr suggested the **quantization rule**, often called Bohr's third postulate:

💡 3. Quantization rule

The electron's angular momentum is restricted to discrete values:

$$m_e v r = n \hbar, \quad n = 1, 2, 3, \dots$$

Here m_e is the mass of the electron, v is electron's speed, r is radius of the stationary orbit, $\hbar = \frac{h}{2\pi}$, and n is a positive integer called a **quantum number**.

³ e.g. Chapter 4 of Eisberg and Resnick. Quantum Physics of Atoms, Molecules, Solids, Nuclei, and Particles. Wiley, New York, 2nd edition, 1985. ISBN 978-0471873730.

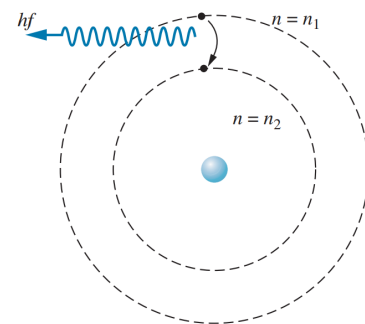


Figure 2: Bohr's model: Transition of electron from stationary state n_1 to stationary state n_2 with emission of a photon. Image [Krane(1996), p.186].

Using classical Newton's forces equation of the electron on a stationary orbit and the quantization rule, Bohr derived an expression for the electron's allowed energies:

$$E_n = -\frac{1}{(4\pi\epsilon_0)^2} \frac{m_e e^4 Z^2}{2\hbar^2 n^2}.$$

The **ground state** corresponds to quantum number $n = 1$, and higher values of n represent excited states.

When an electron jumps from state n to state m , the emitted or absorbed photon frequency is

$$\nu = \frac{E_n}{h} - \frac{E_m}{h}.$$

This matches the structure of the Rydberg formula (1), linking Bohr's theory directly to the observed spectral lines. Equating spectral terms T_n and E_n/h for hydrogen ($Z = 1$), we can express the Rydberg constant R in universal constants $\hbar$, m_e , e and c , known from other experiments:

$$R_\infty = \frac{1}{(4\pi\epsilon_0)^2} \frac{m_e e^4}{4\pi c \hbar^3}. \quad (3)$$

Subscript ∞ here denotes that the value of R is calculated according to the assumption that mass of nucleus is infinitely greater than the mass of electron.

Bohr's model successfully explained the main features of hydrogen's spectrum and gave a physical meaning to the empirical formulas of Balmer and Rydberg. However, it could not account for fine structure, magnetic effects (Zeeman splitting), or multi-electron atoms. These shortcomings were later resolved with the development of quantum mechanics.

Spectral Analysis

Experiments show that many elements have spectral regularities similar to hydrogen. They also form series of spectral lines, but the patterns are usually more complex. According to quantum mechanics, these regularities arise because each atom has a unique set of discrete energy states. The arrangement of these states determines the positions of spectral lines, making every element's spectrum unique — essentially a “fingerprint” of the atom.

This uniqueness makes spectroscopy a powerful tool for **qualitative spectral analysis**. By comparing observed spectral lines with known reference values, one can identify the chemical elements present in a sample.

To study spectra, physicists use special instruments: spectroscopes – for direct visual observation of spectral lines, spectrographs – which

Electron's centripetal acceleration balanced by electrostatic force

$$\frac{m_e v^2}{r} = \frac{e^2 Z}{4\pi\epsilon_0 r^2}$$

$$\nu = \frac{cR}{m^2} - \frac{cR}{n^2} = T_m - T_n$$

Values of constants:

$$\hbar = 1.054\,588\,7 \times 10^{-34} \text{ J} \cdot \text{s},$$

$$m_e = 9.109\,534 \times 10^{-31} \text{ kg},$$

$$e = 1.602\,189\,2 \times 10^{-19} \text{ C},$$

$$c = 299\,792\,458 \text{ m/s},$$

$$\epsilon_0 = 8.854\,187\,819 \times 10^{-12} \text{ F/m}.$$

record spectra photographically, and spectrometers – which use detectors to measure the position and intensity of spectral lines with high precision.

At the heart of each device is a **dispersing element** — either a *prism* or a *diffraction grating*. This element spreads white or mixed light into its component wavelengths. An interesting point is that prisms and gratings disperse light in opposite directions: the prism spreads the colors from red to blue, while the grating spreads them from blue to red (see Figure 3). Why this happens will become clear a bit later.

Prism as a Dispersing Element

When a beam of monochromatic light enters a prism, it bends twice — once upon entering and once upon exiting. The total bending of the beam is described by the deviation angle β , which depends on the prism's **refractive index** $\hat{n}$, the prism's apex angle α and the angle of incidence (Figure 4).

For an equilateral prism ($\alpha = 60^\circ$) the relation between the refractive index and the deviation angle at the position of **minimum deviation** is:

$$\hat{n} = 2 \sin \left(30^\circ + \frac{\beta}{2} \right). \quad (4)$$

The refractive index itself depends on the light's wavelength. This dependence is described empirically by the Cauchy equation [Jenkins and White(1976), p. 479]:

$$\hat{n}(\lambda) = A + \frac{B}{\lambda^2}, \quad (5)$$

where A and B are material constants for the prism. Shorter wavelengths (blue light) are refracted more strongly than longer wavelengths (red light).

Because the refractive properties of the prism glass are not known in advance, the prism must be **calibrated**. Calibration means establishing a relationship between measured deviation angles and known wavelengths using a set of reference spectral lines. Once this relation is determined, the prism can be used to calculate unknown wavelengths.

💡 Calibration basics

Calibration establishes a relation between the measurement device readings and the true physical quantity: we measure a set of data with known values and *calibrate*, i.e., estimate the parameters of a chosen functional dependence between the readings and the quantity. Good to remember the calibration basics. Firstly, the resulting calibration is generally reliable only within

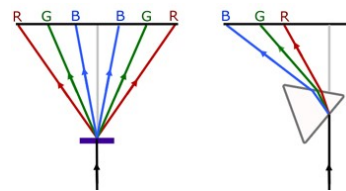


Figure 3: Light diffraction on grating (scheme on left) and dispersion on prism (scheme on right). Note that blue rays, which have a shorter wavelength, are refracted by the prism more strongly than red rays having longer wavelength. On grating (scheme on left) the effect is opposite: red waves are diffracted more than blue. Image: [UTexas](#)

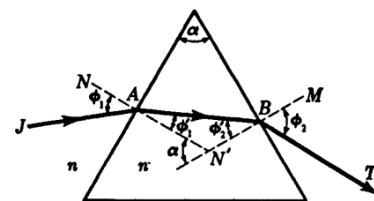


Figure 4: Light refraction on prism. Light ray J falls on prism surface, passes the prism and exits out of second surface as light ray T . The deviation angle between falling ray J and exiting ray T depends on ray J incident angle, refractive index $\hat{n}$ and prism apex angle α . Image: [Jenkins and White(1976), p.31]

the range covered by the observations, so the dataset should span the full working range and subsequent measurements should rely on *interpolation* rather than *extrapolation*. Secondly, choose the simplest adequate model that captures the trend without unnecessary flexibility to escape *overfitting*. Finally, set aside a subset of data not used in fitting for an independent check.

For observed deviation angle β in the equilateral prism with known Cauchy parameters A and B , the wavelength λ can be calculated by combining equations (4) and (5):

$$\lambda = \sqrt{\frac{B}{\hat{n} - A}}, \quad \hat{n} = 2 \sin \left(30^\circ + \frac{\beta}{2} \right), \quad (6)$$

and a calibration equation $\lambda(\beta)$ could be obtained.

Diffraction Grating as a Dispersing Element

A diffraction grating consists of many parallel lines or grooves that cause light to spread into its component wavelengths. The condition for constructive interference, known as the **grating condition**, is

$$n\lambda = d(\sin \theta - \sin i_0),$$

where $n = \pm 1, \pm 2, \dots$ is the diffraction order, λ is the wavelength, d is the spacing between grating lines, i_0 is the angle of light incidence, measured from the grating normal, and θ is the diffraction angle, also measured from the normal.

In practice, the experiment records angles $\beta \equiv \theta - i_0$, measured relative to the incoming beam. After some trigonometric manipulation, the grating condition can be written as:

$$n\lambda = 2d \sin \frac{\beta}{2} \cos \left(i_0 + \frac{\beta}{2} \right), \quad n = \pm 1, \pm 2, \dots \quad (7)$$

Because of the setup, the exact incidence angle i_0 is not measured directly. Instead, it can be determined from the asymmetry of diffraction orders (see Figure 5). If the same spectral line is observed at angles β_+ and β_- for the $+n$ and $-n$ orders, the relation

$$\sin \frac{\beta_+}{2} \cos \left(i_0 + \frac{\beta_+}{2} \right) + \sin \frac{\beta_-}{2} \cos \left(i_0 + \frac{\beta_-}{2} \right) = 0$$

leads to a solution for i_0 :

$$i_0 = \arctan \frac{\sin \beta_+ + \sin \beta_-}{2 - \cos \beta_+ - \cos \beta_-}. \quad (8)$$

This approach allows the experimenter to determine the effective incidence angle indirectly, even when the grating period d and wavelength λ are unknown.

See the textbook
[Jenkins and White(1976), p. 358–360]

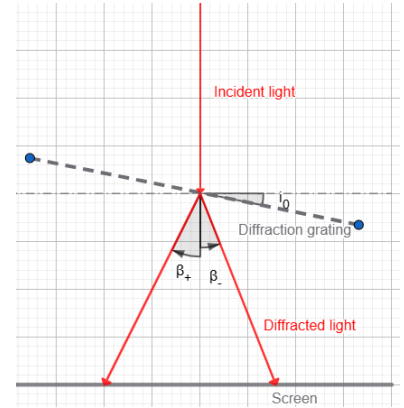


Figure 5: Schematic drawing for incident light falling on diffraction grating plane under an arbitrary non-zero incidence angle i_0 relative to the normal. In this case angles β_+ and β_- for symmetric diffracted orders n_+ and n_- will be not equal.

Performing the Experiment



Safety note

- **High voltage caution!** Please turn off high voltage power supply whenever it is not used.
- **Sensitive optics.** Please do not touch the surfaces of the diffraction grating and other optical elements: lenses, prism. Fingerprints can damage the coatings.

Getting to Know the Workplace

The workplace is equipped with optical spectroscope with digital angle counter, a prism and a diffraction grating (interchangeable as dispersing elements), a set of spectral lamps in one housing, and a high voltage 5 kV power supply.

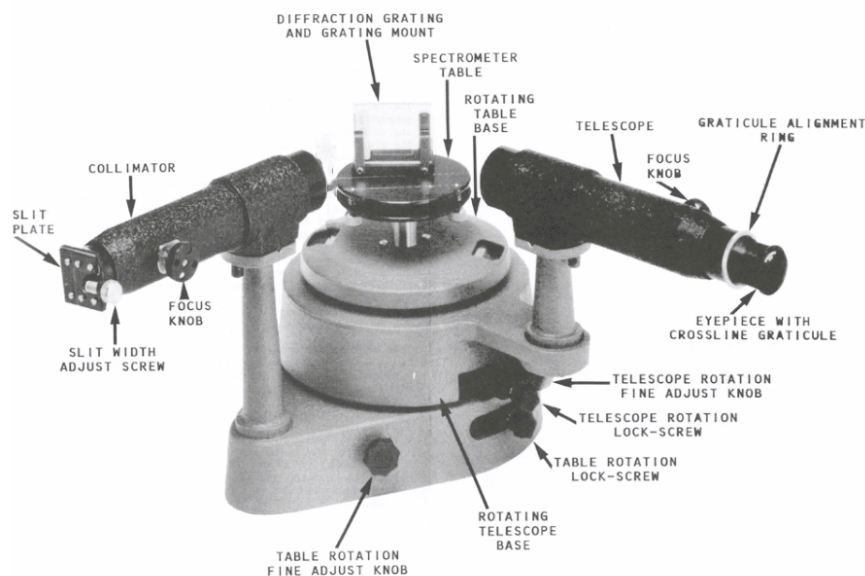


Figure 6: Spectrometer apparatus overview. The manufacturer declares wavelength accuracy measurements up to 0.1 nm when used as a grating spectroscope and up to 2 nm when used as a prism spectroscope. Note the TELESCOPE rotation lock screw and fine-adjust knobs. Image: PASCO.

The optical **spectroscope** (Figure 6) is the classical apparatus designed for a teaching laboratory. The schema of the spectroscope with collimator for light collection and telescope for observations and measurements is shown in Figure 7. The apparatus is equipped with knobs for fixing and fine-tuning the rotation angle of the telescope. Make sure that you understand how to use them. This will be a help when making measurements in the dark.

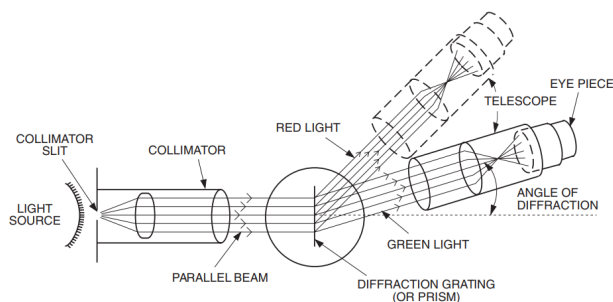


Figure 7: The light rays pass the slit and enter the collimator. Collimator collects the light as a parallel beam and sends to the dispersing element (prism or diffraction grating). The dispersed monochromatic ray can be observed using rotating telescope with eyepiece and measured with angle counter. Image: PASCO.

The digital angle **counter** is a replacement for the original mechanical vernier scales. The digital counter sensor returns 32,000 counts per revolution. The counter's display shows the angle converted to degrees. The display has a button to reset the reading to zero.

The prism and the diffraction grating can be installed on a spectroscopy table as **dispersing element**. The diffraction grating should be fixed on the table with the screw upon installation. The prism should not be fixed to allow minimal deviation angle tuning.

The set of **spectral lamps** is equipped with mercury, helium and hydrogen lamps. The sliding protective plank opens and closes the lamp. Each lamp has its own plug to the power supply.

The high voltage 5 kV **power supply** has ON/OFF switch and HYDROGEN switch. This switch is used to make the spectral lamp light brighter.

The experiment is performed in a darkened room. A **backlight lamp** under the table does not interfere with observations and can be used for writing notes or to backlight the crosshair in the eyepiece. The lamp switch is located on the right side below the table surface.

Setting the Apparatus

1. PLUG IN the light source lamp and TURN ON the high voltage power supply. Shift the plank on the lamp housing to open the lamp and ensure that you see the light source.
2. Gently ALIGN the apparatus collimator straight against the light source. FIX the table rotation lock-screw. Turn aside the telescope tube to ensure that you see bright vertical line on the second lens of the collimator.
3. INSTALL the dispersing element: prism or grating.
4. ALIGN the telescope with the collimator (see Figure 8).
5. Make sure by adjusting SLIT width screw and FOCUS knobs that the light source is seen in eyepiece as narrow sharp vertical line.

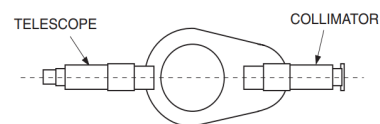


Figure 8: Telescope aligned directly opposite the collimator. Image: PASCO.

Make sure that you see the crosshair. Crosshair eyepiece is adjustable: shift it along optical axis to see the crosshair in focus.

6. Fine-tune the telescope rotation angle with LOCK screw and FINE ADJUST knob to make sure that the sharp narrow vertical line of the light source coincides with crosshair you see in the eyepiece.
7. SET the digital angle counter to zero.

Measurements with Prism

1. INSTALL the prism on the spectrometer table as shown on the Figure 9. Do not fix the prism table: you will need to swing it when finding the minimal deviation angles.
2. TURN ON the **helium** lamp and SET the apparatus to zero position as described above.
3. ROTATE the telescope counter-clockwise until you see the lamp spectral lines. They are expected to be seen somewhere at angles $40^\circ - 60^\circ$.
4. IDENTIFY each spectral line. Refer to NIST database or see the figures in [Appendix. Spectrum lines wavelengths](#). You should see seven strong spectral lines, from blue to red, and some weaker lines.
5. Choose a spectral line and focus on it. Identify its wavelength. SWING gently the prism table. You will see the line is shifting left (to smaller angles) or right (to greater angles). Find the minimal (leftmost) deviation angle of the line.
6. Fine-tune the telescope position. Estimate the observation uncertainty. RECORD the observation in the Table 1.
7. REPEAT the measurements for all helium spectral lines identified.
8. TURN OFF the high voltage supply and unplug the lamp.
9. Turn on the **mercury** lamp, align the apparatus and REPEAT the measurements for the mercury strong green line. Write it down in the table. You will use this measurement to test the calibration.
10. Build weighted linear regression model ([Taylor(1996), Ch.8]) to find calibration parameters A and B with their uncertainties using helium observation data, prism equation (4) and Cauchy empirical equation (5).
11. Test your model on mercury green line observation using eq. (6). The wavelength λ true value should lie within uncertainty interval.

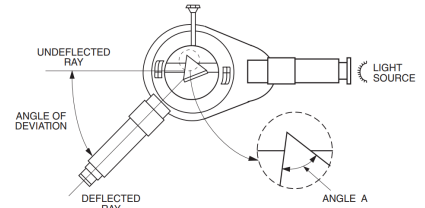


Figure 9: Prism orientation, view from top. Image: PASCO.

Line ID	β , deg	$\delta\beta$, deg
Red 706	48.000	0.033
Red 668	..	..
..	..	..

Table 1: Sample table for prism measurements

$$\hat{n} = 2 \sin \left(30^\circ + \frac{\beta}{2} \right), \hat{n} = A + \frac{B}{\lambda^2}.$$

$$\lambda = \sqrt{\frac{B}{\hat{n} - A}}$$

12. Repeat the measurements for **hydrogen** spectrum lines. You should see three or four spectrum lines. Identify the lines using NIST database or by referring to the Appendix.
13. Apply the model and calculate the hydrogen spectrum wavelengths with their uncertainties.

Measurements with Diffraction Grating

In this part of the experiment we perform measurements immediately with the hydrogen spectrum lamp, assuming that the diffraction grating parameter d is known and we accept its nominal value. When d is unknown, calibration on the known wavelengths of helium and mercury should be performed before hydrogen measurements.

1. TURN ON the **hydrogen** lamp and ALIGN the apparatus against the lamp as described above. SET the digital angle counter to zero.
2. INSTALL the diffraction grating and fix it with the screw. **It is important not to move the apparatus or the grating after set up, otherwise the results will be inconsistent.**
3. Observe light diffraction by rotating telescope clockwise and counter-clockwise. In both directions you should see spectrum lines of several orders — the first orders $n = \pm 1$ are the brightest, the others are less and less bright. You may need to increase the light brightness by turning on the HYDROGEN switch on the power supply for short time. In each order you should see lines of the hydrogen spectrum. Identify them referring to NIST database or with the Figure 13 in the Appendix.
4. Choose a spectral line and focus on it. IDENTIFY the line's wavelength λ and the order n . Fine-tune the telescope rotation angle with LOCK screw and FINE ADJUST knob to make sure that the sharp narrow spectral line coincides with the crosshair. Measure the deviation angle β . Estimate its uncertainty. RECORD the observation to the Table 2.
5. Proceed with all spectral lines seen, both positive and negative orders.
6. TURN OFF the high voltage power supply and disconnect lamp.
7. For all pairs of the same spectral color of order $\pm n$ calculate i_0 as in eq.(8). Use weighted average [Taylor(1996), Ch. 7] to estimate value of i_0 and its uncertainty.

Line ID	n	β, deg	$\delta\beta, \text{deg}$
Red 656	1	9.045	± 0.011
..	..	..	..

Table 2: Sample table for measurements with a diffraction grating.

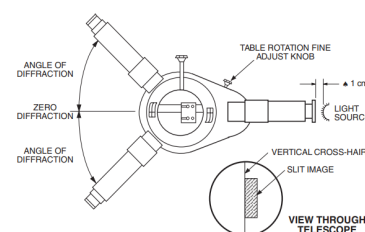


Figure 10: Measuring diffraction angle. Image: PASCO

$$i_0 = \arctan \frac{\sin \beta_+ + \sin \beta_-}{2 - \cos \beta_+ - \cos \beta_-}.$$

8. Use equation (7) for all measured orders n and angles β_n to calculate wavelengths. Again, use error propagation and weighted average to estimate values with their uncertainties.

$$n\lambda = 2d \sin \frac{\beta_n}{2} \cos \left(i_0 + \frac{\beta_n}{2} \right)$$

Data Analysis

1. Having hydrogen Balmer series wavelengths λ measured with prism and diffraction grating spectroscopy methods, identify their respective quantum numbers n and calculate Rydberg's constant R from empiric equation (1) using weighted averages for uncertainty estimation. Note that n as diffraction order, referred above, and quantum number n here are not the same.
2. Compare measured Rydberg's constant R with theoretical expectation (3) derived from Bohr's postulates.

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

$$R_\infty = \frac{1}{(4\pi\epsilon_0)^2} \frac{m_e e^4}{4\pi c \hbar^3}$$

References

- [Bohr(1913)] Niels Bohr. On the constitution of atoms and molecules. part i. *Philosophical Magazine*, 26(151): 1–25, 1913. doi: 10.1080/14786441308634955.
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- [Krane(1996)] Kenneth S. Krane. *Modern Physics*. Wiley, New York, 2nd edition, 1996. ISBN 978-0471828723.
- [Taylor(1996)] John R. Taylor. *An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements*. University Science Books, 1996. ISBN 093570275X.

Acknowledgments

The theoretical part of this document is inspired by the laboratory experiment manual [Studying the Atomic Hydrogen Spectrum](#), written by Associate Professor G.E.Pustovalov, Moscow State University, Physics Dept, 2012 (in Russian).

Spectrometer apparatus images and setting instructions adapted from PASCO Student spectrometer manual (model sp-9268a) [Instruction Manual and Experiment Guide](#).

Appendix. Spectrum lines wavelengths

Here are shown the results of spectral lamp intensity measurements performed on the laboratory lamps set with a digital spectrograph *Avantes*. Note that the the figures are for illustration only. Follow to [NIST](#) reference database for accurate wavelengths values.

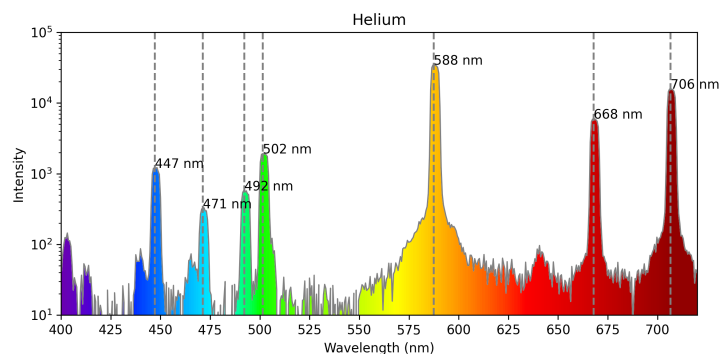


Figure 11: Helium lamp spectrum measurements. Strong lines of He spectrum occur at 447.1 nm (blue), 471.3 nm (weaker blue), 492.1 nm (weaker green), 501.5 nm (green), 587.5 nm (strong yellow), 667.9 nm (red), 706.5 nm (red). Weak lines are 402.6 nm and 414.4 nm (violet), 438.8 nm and 468.5 nm (blue).

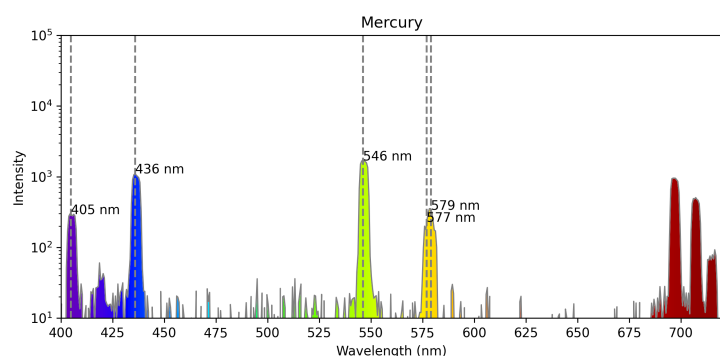


Figure 12: Mercury lamp spectrum measurements. Strong line of Hg spectrum occurs at 435.8 nm (strongest blue line), 546.1 nm (strong green line), yellow-orange: 577.0 nm and 579.1 nm (a pair of strong lines). Weaker blue line: 404.7 nm. Note that red lines at 697 nm and 707 nm are not of Hg spectrum. These red lines belong to argon, which is used as starter gas.

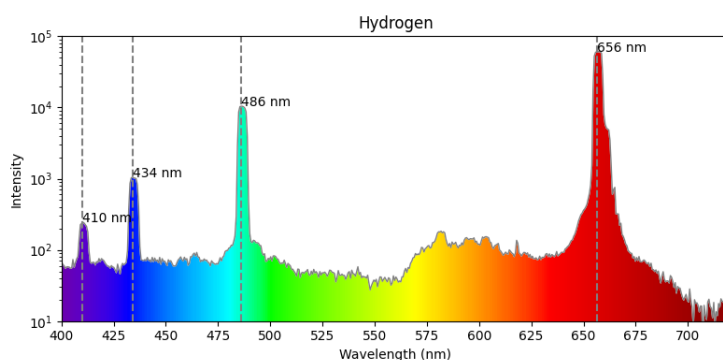


Figure 13: Hydrogen lamp spectrum measurements. Strong lines are 656.3 nm (red), 486.1 nm (blue-green), 434.0 nm (blue). Weaker line is 410.2 nm (weak violet). These lines are emitted by an electrons transition to the $n = 2$ energy level from higher levels ($n = 3, 4, 5$, and 6 respectively).